

$$m_l = 3.33 \times 10^5$$

$$= \mathbf{0.440 \text{ kg}}$$

4(a)(i)

Diffraction of light refers to bending or spreading of light through an aperture or around an obstacle of dimensions comparable to the wavelength.

4(a)(ii)

Diffraction grating formula: $\frac{1}{N} \sin \theta = n\lambda$

Or $\sin \theta = Nn\lambda$

To see images on the screen, $\theta < 90^\circ$ or $\sin \theta < 1$

$$\therefore Nn\lambda < 1 \text{ or } n < \frac{1}{N\lambda}$$

The maximum order of lines $n_{\max}$ is the largest value of n expressed as an integer.

$$4(b)(i) \text{ Gradient} \quad \quad \quad = (0.44 - 0.28)/[(550 - 350) \times 10^{-9}] \quad \quad = \mathbf{8.0 \times 10^5 \text{ m}^{-1}}$$

4(b)(ii) Diffraction Grating formula: $n\lambda = d \sin \theta$, or $2\lambda = d \sin \theta$

Plotting $\sin \theta$ vs λ , the gradient $= 2/d$ $8.0 \times 10^5 = 2/d \Rightarrow \mathbf{d = 2.5 \times 10^{-6} \text{ m}}$

4(b)(iii) $\lambda = d \sin \theta$, plotting $\sin \theta$ vs λ gives gradient $= 1/d$

Hence a straight line with half the original gradient will be obtained.

The line will have the same y intercept *at the origin of the graph*.

Q5(a)(i)

For a floating object, $F_{\text{up}} = F_{\text{down}}$